

Week 1: Vectors – Supplementary Notes

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These notes support the Week 1 lecture. They are not covered in Young and Freedman, Chapter 1, and they are not required for the Week 1 lecture problems. Read them after the lecture.

1 Linear independence of the unit vectors

In the lecture we stated that the mutually perpendicular unit vectors $\hat{i}$, $\hat{j}$ and $\hat{k}$ provide three independent directions, so a vector is uniquely specified by its x -, y - and z -components. This section states that idea precisely.

Definition: linear independence

A set of vectors $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n$ is **linearly independent** if the only way to combine them to give the zero vector is with every coefficient equal to zero:

$$c_1\vec{u}_1 + c_2\vec{u}_2 + \dots + c_n\vec{u}_n = \vec{0} \implies c_1 = c_2 = \dots = c_n = 0.$$

The unit vectors are linearly independent. Suppose $a\hat{i} + b\hat{j} + c\hat{k} = \vec{0}$. Take the dot product of both sides with $\hat{i}$. Because $\hat{i} \cdot \hat{i} = 1$ and $\hat{i} \cdot \hat{j} = \hat{i} \cdot \hat{k} = 0$, this gives $a = 0$. Dotting with $\hat{j}$ and $\hat{k}$ in the same way gives $b = 0$ and $c = 0$.

Geometrically, no combination of $\hat{i}$ and $\hat{j}$ can produce $\hat{k}$, because any combination $a\hat{i} + b\hat{j}$ lies in the xy -plane, while $\hat{k}$ is perpendicular to that plane.

Consequence: components are unique

If $\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$ and also $\vec{A} = A'_x\hat{i} + A'_y\hat{j} + A'_z\hat{k}$, then subtracting gives

$$(A_x - A'_x)\hat{i} + (A_y - A'_y)\hat{j} + (A_z - A'_z)\hat{k} = \vec{0},$$

so $A_x = A'_x$, $A_y = A'_y$ and $A_z = A'_z$. Hence two vectors are equal if and only if their corresponding components are equal:

$$\vec{A} = \vec{B} \iff A_x = B_x, \quad A_y = B_y, \quad A_z = B_z.$$

This is why a single vector equation can be split into separate x , y and z equations, a technique used throughout mechanics (for example, when resolving Newton's second law into components).

2 Extension: Direction Angles in 3D

In two dimensions a single angle θ specifies a direction. In three dimensions we can instead give the three angles α , β and γ that the vector $\vec{A}$ makes with the positive x -, y - and z -axes. Their cosines are called the **direction cosines**.

Using the dot product with each unit vector, $\vec{A} \cdot \hat{i} = A \cos \alpha$ and also $\vec{A} \cdot \hat{i} = A_x$. Therefore:

Rule: direction cosines

$$\cos \alpha = \frac{\vec{A} \cdot \hat{i}}{A} = \frac{A_x}{A}, \quad \cos \beta = \frac{\vec{A} \cdot \hat{j}}{A} = \frac{A_y}{A}, \quad \cos \gamma = \frac{\vec{A} \cdot \hat{k}}{A} = \frac{A_z}{A}$$

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

The identity follows directly: $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = (A_x^2 + A_y^2 + A_z^2)/A^2 = 1$. The direction cosines are also the components of the unit vector along $\vec{A}$:

$$\hat{A} = \frac{\vec{A}}{A} = \cos \alpha \hat{i} + \cos \beta \hat{j} + \cos \gamma \hat{k}.$$

Worked example

Find the direction angles of $\vec{A} = \hat{i} + 9\hat{j} + 2\hat{k}$ (the vector used in Contact 3), and check the identity.

Magnitude: $A = \sqrt{1^2 + 9^2 + 2^2} = \sqrt{86} = 9.27$.

Direction angles:

$$\alpha = \cos^{-1} \frac{1}{9.27} = 83.8^\circ, \quad \beta = \cos^{-1} \frac{9}{9.27} = 13.9^\circ, \quad \gamma = \cos^{-1} \frac{2}{9.27} = 77.5^\circ.$$

Check: $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = \frac{1 + 81 + 4}{86} = 1$.

Evaluate: β is small, so $\vec{A}$ points mostly along $+y$, which agrees with its large y -component.

Use. Direction angles provide a compact way of describing the orientation of a vector in three dimensions.

Exercise. Find the direction angles of $\vec{B} = -5\hat{i} + \hat{j} - \hat{k}$. (Answer: $\alpha = 164.2^\circ, \beta = 78.9^\circ, \gamma = 101.1^\circ$.)